Energy Consumption Analysis Using PCA and K-Means

Graphs and simple explanations

About this project

For this project we looked at a dataset of weekly energy use. It has **321 series** (rows) and **156 weekly columns**. We wanted to see what patterns are in the data and whether the series can be grouped into clusters.

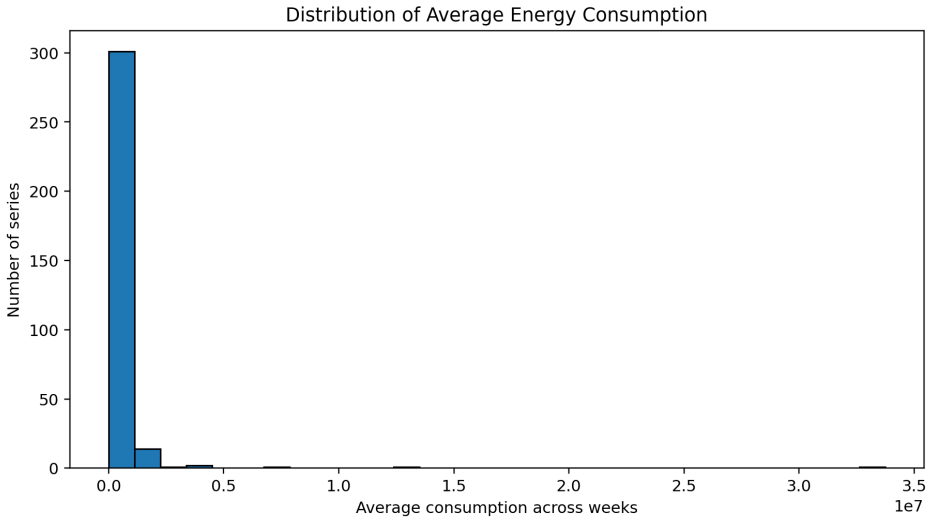
What we did

First, we changed the weekly values to numbers and filled any missing values with the median of that column. Next, we **standardized** the data so that no week counts more just because its numbers are bigger. Then we used **PCA** to reduce the data to two dimensions, and **K-Means** to group the series into clusters.

Main results

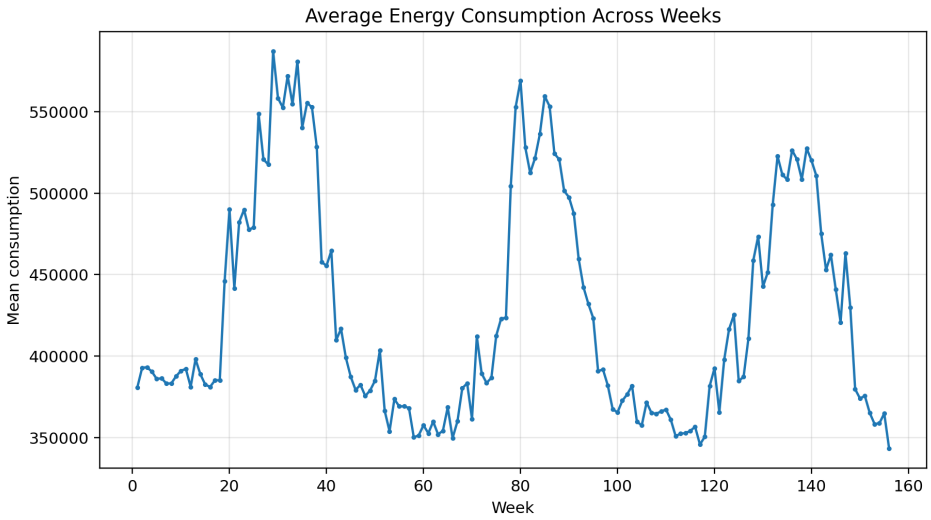
The first two PCA components explain about **99.9%** of the variance. we used **K = 3** for K-Means, and the silhouette score is **0.957**. A score close to 1 usually means the clusters are well separated, but as shown below, this is partly because of a few very large series.

1. Distribution of Average Energy Consumption



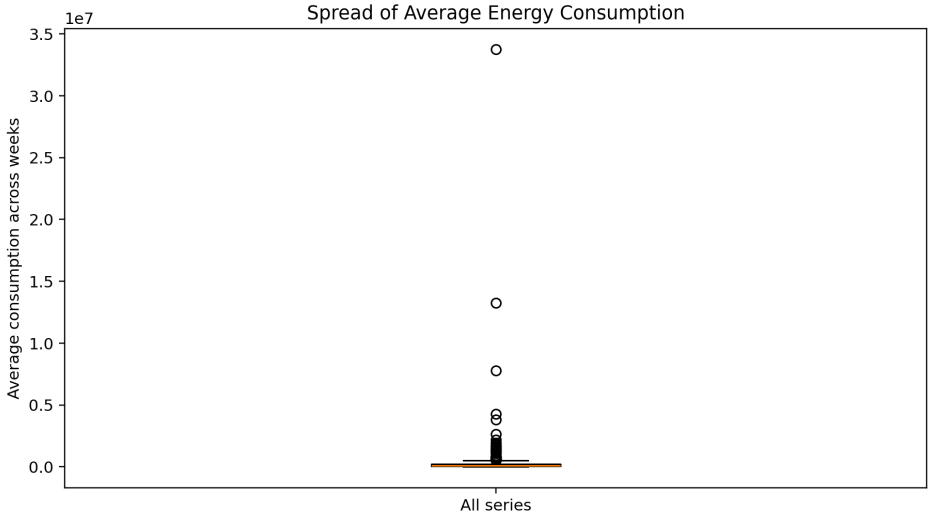
This histogram shows how the average weekly consumption is spread out across the 321 series. Almost all of them (about 300) are in the very first bar, so most series use a small amount of energy. A few series use a lot more, and the biggest one is around 3.4×10^7. These big ones are so rare that their bars are hard to see. This means the data is very **skewed**.

2. Average Energy Consumption Across Weeks



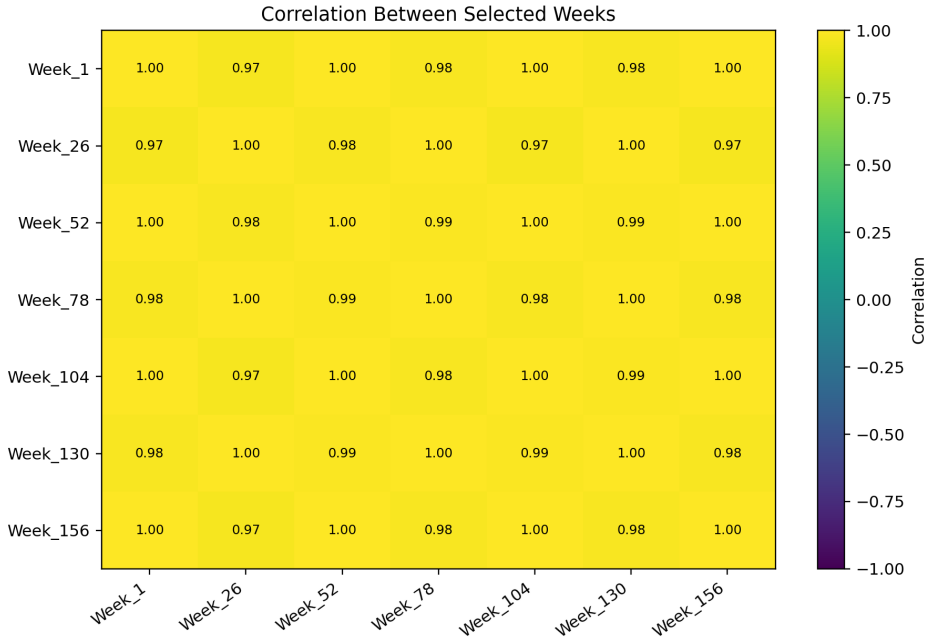
This graph shows the mean consumption for each week. The pattern repeats three times, with peaks near weeks 30, 80 and 135. The peaks are about 50 weeks apart, so it looks like a **yearly cycle** (maybe winter or summer use). During a peak the average goes up to around 550,000 to 585,000, and in between it goes down to about 350,000. The third peak is a bit lower than the first two, but with only three cycles we can't be sure that is a real trend.

3. Spread of Average Energy Consumption



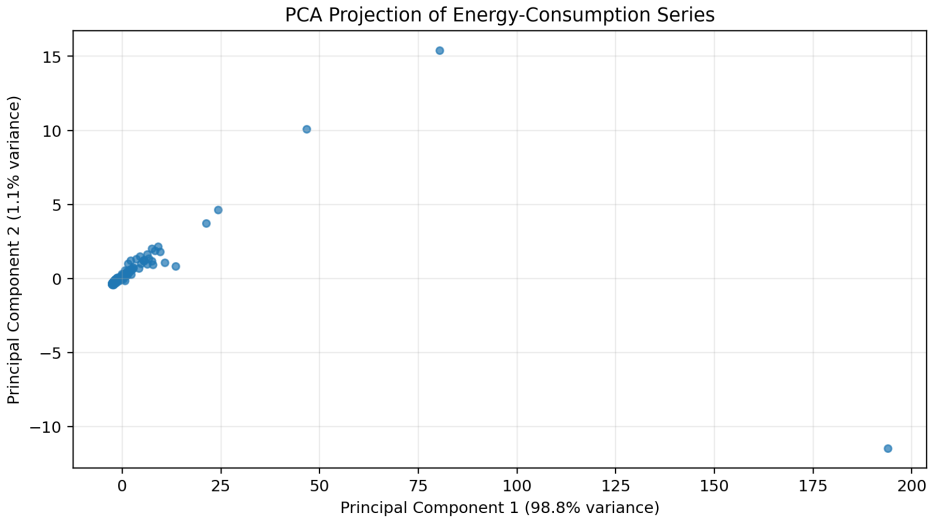
The boxplot shows the median, the spread and the outliers. The box is squashed at the bottom because most values are small. There are lots of outlier dots above it, and the highest ones are around 3.4×10^7, 1.3×10^7 and 7.7×10^6. This agrees with the histogram: a small number of series are much bigger than the rest.

4. Correlation Between Selected Weeks



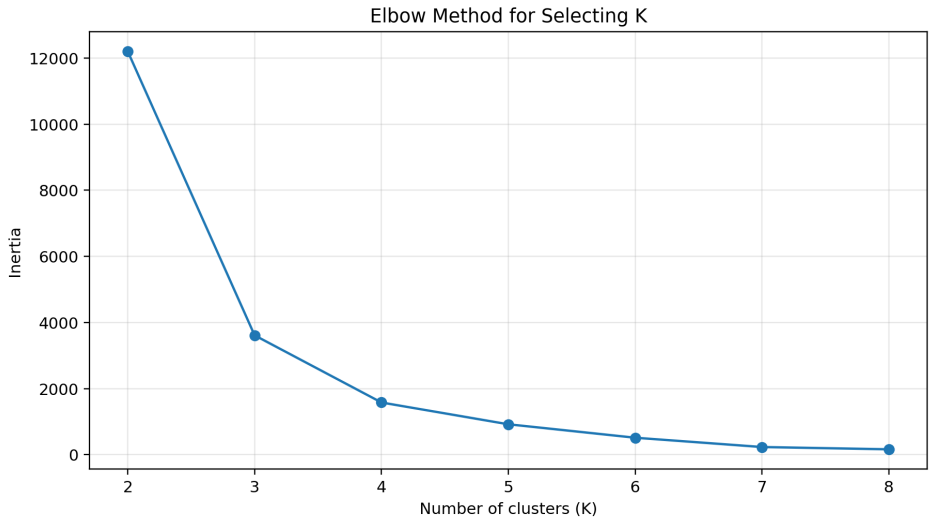
We picked seven weeks (1, 26, 52, 78, 104, 130 and 156) and checked how they relate to each other. All the correlations are between 0.97 and 1.00, which is almost perfect. So if a series has high usage in one week, it usually has high usage in the other weeks too. The main difference between series is their **overall size**, not when they use energy. This is also why PCA can squeeze the data so well.

5. PCA Projection



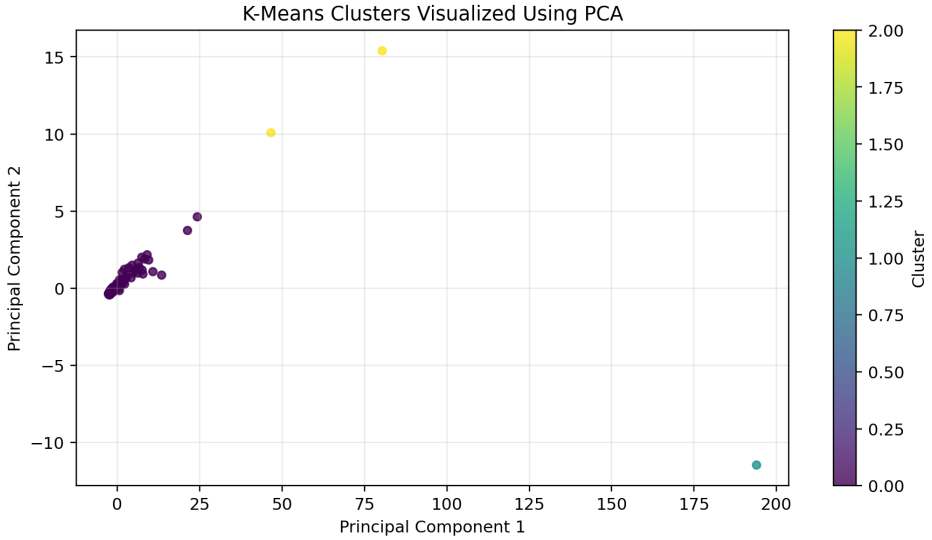
PCA reduces the 156 weekly columns to just 2 dimensions so we can plot them. The first component (PC1) explains 98.8% of the variance and the second (PC2) explains 1.1%, so together they cover about 99.9%. Most points are bunched up near the origin, and a few are far away. One point is at about 194 on PC1. PC1 probably represents the overall consumption level, which fits with the correlation results.

6. Elbow Method



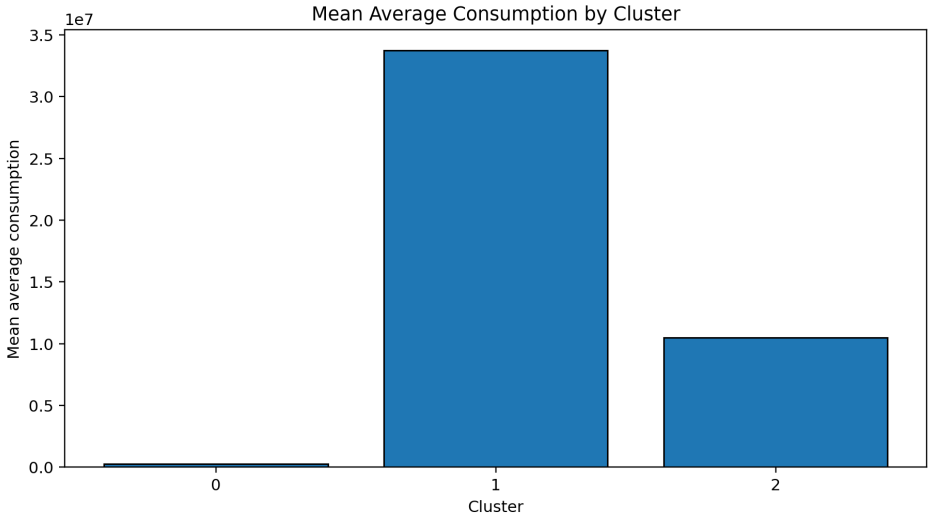
We ran K-Means with K from 2 to 8 and plotted the inertia (how spread out the points are inside their clusters). The line drops quickly from K = 2 (about 12,000) to K = 3 (about 3,600) and then flattens out. The bend is at K = 3, so we chose **3 clusters**. K = 4 might also work, but the gain after 3 is small.

7. K-Means Clusters Visualized Using PCA



Here each dot is one series, coloured by its cluster. One big cluster (cluster 0, dark purple) holds almost all the series. The other two clusters are very small: cluster 1 is a single series far to the right, and cluster 2 is two series near the top. So K-Means mostly pulled out the extreme cases instead of making groups of similar size.

8. Mean Average Consumption by Cluster



This bar chart compares the mean consumption of each cluster. Cluster 1 is the highest at about 3.4 x 10^7, cluster 2 is about 1.05 x 10^7, and cluster 0 is almost zero on this scale. So the clusters are basically sorted by how much energy the series use.

Conclusion

The data has a clear yearly pattern, and the series mostly differ in how much energy they use, not in the shape of their usage. The clustering score looks great (0.957), but the clusters are very uneven: one has almost all the series and the other two have just three series between them. This happens because a few extreme outliers dominate the data.

If we did this again, we would use a **log transform** or treat the outliers separately. That way the clusters could show real differences between the other 318 or so series.